

AI-Driven Differential Diagnosis: Leveraging RAG and LLMs in Intelligent Healthcare Systems

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Abstract—Artificial Intelligence is playing an increasingly prominent role in healthcare. For this integration to continue evolving effectively, professionals in the field, including doctors, psychologists, and nurses, must trust the technology. In the domain of Large Language Models, this trust requires transparency in the sources of information, ensuring they are verifiable and reviewed. This paper presents an intelligent diagnostic assistant designed to support differential diagnosis and disease comparison, leveraging an LLM enriched with the Retrieval-Augmented Generation technique to address trust-related challenges. The knowledge base for our system is PubMed, a biomedical article aggregator, which supplies relevant articles on the considered disorders, as well as evidence supporting or refuting potential links between them. Retrieval-Augmented Generation empowers the model to incorporate external knowledge tailored to the specific question while providing citations for its statements. These citations not only enhance the trustworthiness of the answers but also enable practitioners to explore the referenced articles in greater detail. We developed a prototype of our system, and conducted an evaluation through questionnaires administered to 12 professional psychologists. Results show that our system compares favorably with state of the art Large Language Models in providing the relevant information to distinguish between two diseases, and that the provided details are useful for supporting differential diagnosis.

Index Terms—Pervasive Healthcare, Retrieval-Augmented Generation, Trust and Transparency in AI, Biomedical Knowledge Integration, Human-Computer Interaction in Medicine

I. INTRODUCTION

Artificial Intelligence (AI) is becoming increasingly prominent in medicine, transforming clinical practice through pervasive healthcare; i.e., the integration of new digital technologies to enhance various aspects of healthcare [1]. Indeed, AI is already widely applied across various medical fields, from robot-assisted surgeries [2] to protein complex prediction in research [3], and diagnosis of neurodegenerative diseases [4]. Despite its potential, many professionals in the medical fields remain hesitant to adopt AI solutions, often due to concerns over transparency, particularly when Large Language Models (LLMs) are employed [5].

LLMs are AI systems trained on vast datasets, including articles, books, and internet content, enabling them to perform several natural language processing tasks [6]. While large datasets are essential for training, it is recognized that in many cases the quality of data is more critical than quantity [7]. Reliable, verifiable sources are crucial to building trust, especially

in the medical field, as is providing users with transparency about data provenance. However, it is well known that current LLMs can suffer from hallucinations, generating convincingly false or nonsensical information [8]. Of course, in healthcare, such errors can discourage professionals from relying on AI for tasks like diagnosis [9]. The black-box nature of LLMs, where training data and architecture remain undisclosed, limits user trust. While domain experts are expected to verify responses, errors are not always easy to identify, especially in specialized areas lacking robust training data. For this reason, it is necessary to devise innovative AI techniques for transparency and explainability in pervasive healthcare [10].

Recent advancements, such as MedPALM 2 [11] and BioBERT [12], rely on fine-tuned models using medical-specific data for better domain accuracy. Another promising technique is Retrieval-Augmented Generation (RAG), which integrates external knowledge sources to reduce hallucinations and improve transparency. RAG is particularly useful for handling tasks requiring domain-specific knowledge, especially in the medical field, as it provides more reliable answers despite limited training data [13].

RAG-based systems have shown better performance than traditional LLMs, especially in cases with sparse data, and can outperform larger models when using external knowledge [14]. This approach does not require extensive computing resources, unlike fine-tuning methods. The Llama 2 family, particularly the Llama 2-Chat model, has been adapted for medical applications, benefiting from its open-source nature and fine-tuning on biomedical corpora [15]. Solutions like PMC-LLaMA [16] inject biomedical knowledge to enhance model performance.

Additionally, Wu et al.'s MedGraphRAG integrates knowledge graphs with RAG to improve reasoning, offering greater accuracy in medical questions by linking entities and clarifying relationships. Smaller models like Llama 2 8B have shown impressive results using datasets like MedC-K and MIMIC-IV, which provide de-identified patient data [17]. These datasets hold potential for enhancing AI-driven diagnosis, although careful information extraction is needed for reliability. However, to the best of our knowledge, none of the existing solutions are specifically addressed to support *differential diagnosis*; i.e., the process of distinguishing between diseases with similar symptoms to determine the most likely cause of a patient's condition [18].

In this paper, we propose an AI solution designed to act as an intelligent healthcare assistant during differential diagnosis. Our proposed AI assistant supports the practitioner by retrieving and presenting relevant, evidence-based information. It uses peer-reviewed medical literature to provide insights into possible diagnoses, supporting the physician in evaluating the likelihood of each condition. Our AI tool employs a chatbot interface, designed with domain-specific knowledge to assist doctors, and ensure reliability by focusing on well-defined, authoritative medical contexts. The proposed solution addresses trust issues by sourcing information from peer-reviewed articles, which offer a robust foundation for providing both diagnostic and research assistance.

Technically, the work presented in this paper integrates the RAG technique into a LLM, pre-trained without medical fine-tuning, to assist physicians in differential diagnosis. As being sourced from the trusted database PubMed, a free biomedical literature aggregator maintained by the National Center for Biotechnology Information (NCBI) [19], our approach aims to reduce hallucinations and therefore to increase the acceptance by medical experts.

We implemented a working prototype of our system using Llama 2, and experimentally compared it with state of the art solutions relying on more powerful LLMs, such as ChatGPT 4o, and with a LLM empowered by a standard knowledge injection method. To evaluate the system, a group of 12 psychologists was presented with differential diagnosis cases, with statements generated by three methods: (A) a Llama LLM enriched with knowledge injection, (B) ChatGPT 4o, and (C) the RAG-enhanced LLM method presented in this work. Respondents rated the statements via questionnaires, assessing their utility and trust for diagnosis. Achieved results show broad agreement that this type of technology is useful, with our method emerging as the preferred solution overall. As expected, ChatGPT 4o, being trained on a much larger set of text, scored higher for response clarity than Llama 2 solutions, while the knowledge-injected LLM outperformed the other methods in grading the similarity between disorders.

To summarize, this paper provides the following contributions:

- We designed a novel method to extract context from PubMed, a biomedical literature aggregator, with a focus on articles that evaluate links between two disorders;
- We applied RAG to a pre-trained LLM to assist experts in the field of differential diagnosis;
- We conducted a comparative evaluation of our RAG-enhanced LLM against a knowledge-injected pre-trained LLM and a powerful ChatGPT 4o model. This evaluation was based on responses to differential diagnosis cases provided by 12 professional psychologists through structured questionnaires.

The rest of the paper is structured as follows. Section II sets out the solution to solve the problems outlined in the introduction. In that section, the concept of RAG and how it was integrated is explored. Section III presents the technologies used in the concrete implementation. Subsequently, the workflow of

the complete system is explained through an example diagram. Section IV is dedicated to the presentation of the experimental evaluation and the discussion of the achieved results. Section V concludes the paper by summarizing the work carried out, highlighting open problems and future developments.

II. PROPOSED SOLUTION

In this section, we present the rationale of our proposed solution for AI supported differential diagnosis.

A. Requirements and rationale

The primary requirement to be addressed by our system is information reliability, adhering to the principle that “Misinformation can be worse than no information at all” [13]. Achieving this requirement requires two components: (i) a reliable knowledge source, and (ii) a highly deterministic model that does not provide answers with unverified information.

To address this challenge, our approach is to augment a LLM knowledge base with external knowledge acquired from PubMed, a reliable biomedical database, which is an ideal source due to its extensive collection of indexed peer-reviewed articles. PubMed is freely accessible and offers API integration for article searching [20].

B. LLMs and RAG

As introduced before, a major issue for the use of LLMs in healthcare is the occurrence of hallucinations. Fine-tuning a large LLM with medical data can address this issue, but it requires significant computational effort, and is not easily scalable to the continuous addition of new information. In contrast, the RAG approach enhances LLMs by incorporating specific external knowledge at runtime. RAG allows LLMs to access non-parametric memory, stored as vectors representing documents, which are retrieved based on the input query [21]. The retrieved information is incorporated into the model’s response. This approach involves a pre-trained model, a retriever, and a vector database.

To ensure efficiency, the retrieved context is split into chunks of a fixed size. While too large chunks introduce noise, smaller ones may lack context. Recursive chunking helps overcome this problem, splitting the text until the optimal size is reached. Key advantages of RAG consist in transparency and explainability, since the sources of the retrieved information are always traceable. To avoid hallucinations, the model must be guided to follow only the provided context, limiting sampling temperature and new tokens.

C. Architecture and algorithms

Figure 1 illustrates our system architecture.

According to the queried symptoms and diseases, our system automatically queries the PubMed Knowledge base to extract the relevant documents. The document text is extracted from the paper abstracts, divided in chunks, and embedded. In order to feed our RAG technique, we use non-parametric memory, i.e., not used during training, stored as a dense vector index accessible with a pre-trained retriever [21]. The vector

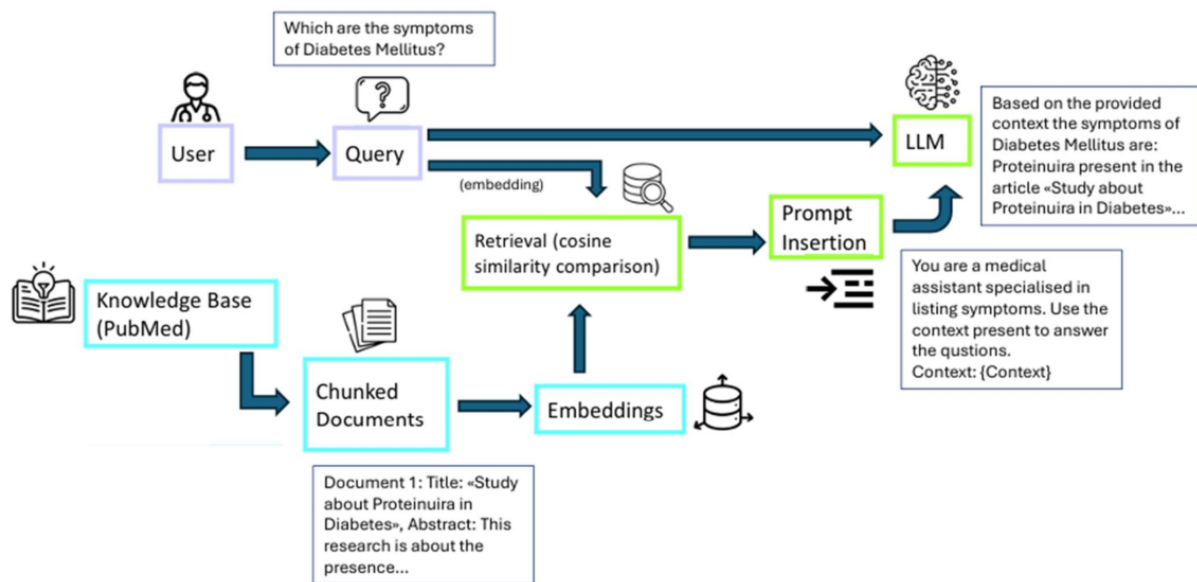


Fig. 1. A graphic representation of our technique.

relative to the input query (e.g., a query about the symptoms of diabetes mellitus) is compared with the vectors of the most similar documents within the index and the top- k returned by the retriever module are selected. The obtained context is coherently inserted into the prompt of the pre-trained model that, in addition to the parametric memory, will use this new knowledge to respond to user input.

In our system, the desired context is often too lengthy to be processed in its entirety, necessitating a subdivision into smaller, manageable chunks. To achieve this, we utilize fixed-size chunking, where the dimensions of the chunks are carefully calibrated. Chunks that are too large introduce noise and incur higher computational costs, while chunks that are too small fail to capture sufficient context, reducing their utility. Given the variable length of article abstracts, some chunks may lack consistency with the rest of the document, isolating only highly specific sentences. To mitigate this issue, our algorithm incorporates overlapping characters between adjacent chunks, enhancing their consistency and ensuring smoother transitions.

However, the unstructured nature of PubMed texts poses additional challenges. For this reason, we adopt a recursive chunking technique. Our method iteratively splits the text until the optimal chunk size is achieved. If the optimal size is not reached on the first attempt, the process is repeated recursively. While this approach is computationally more expensive than fixed-size chunking, it consistently delivers superior results and ensures that the algorithm maintains a high level of performance.

Once the documents are retrieved and divided into chunks, they undergo vectorization using embedding representations. Word embeddings are dense, fixed-length vectors that capture the semantic meaning of words, constructed based on word co-occurrence statistics, as per the distributional hypothesis [22]. The user input is similarly transformed into an embedding

representation and compared against the stored vectors in the database using a similarity metric, such as cosine similarity. This comparison ensures that the most relevant information is accurately retrieved and provided to the model as input, thereby enhancing the precision and reliability of the generated response.

When a new query is issued, our algorithm identifies the top- k most relevant documents from the knowledge base and inserts them into the prompt. This ensures that the model has access to relevant information sourced from reliable and citable references. Our system also explicitly displays the reference of the documents used to generate the answer, in order to provide transparency.

However, merely incorporating additional context is insufficient to prevent hallucinations in the language model. For this reason, our algorithm employs a template prompt that enforces strict adherence to the provided context. Additionally, the sampling temperature is kept at a low level, the number of new tokens in the output is minimized, and the size of the context window is carefully limited to constrain the model's generative capability.

III. SYSTEM IMPLEMENTATION

This section presents the concrete implementation of the solution described in the previous section tailored to support differential diagnosis, highlighting the specific technologies that we adopted for the considered task.

A. Technologies

Overall, our system takes as input a pair of diseases to be compared for differential diagnosis. From this pair, a context is constructed using the abstracts of articles in PubMed that match the considered diseases. The text with concatenated abstracts is divided into chunks, vectorized and stored in a

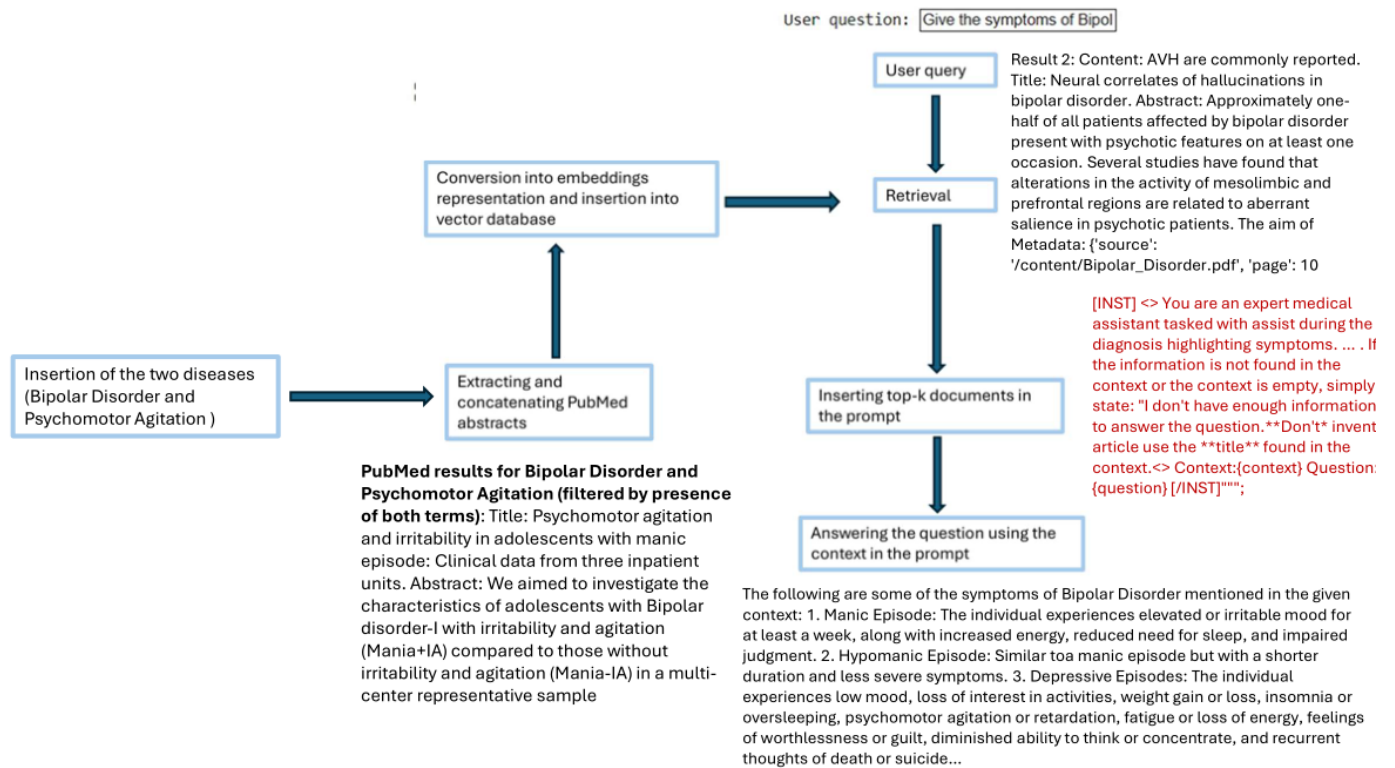


Fig. 2. System workflow diagram with a concrete example.

vector database. Its contents is compared with the vector derived from the user query to choose the top- k chunks to be used for generating the comparison. The template with the appropriate prompt and enriched by the new context is used to answer the user's query using the Llama 2 7B Chat LLM.

In order to connect to PubMed without downloading the database locally, we use the E-utilities API of the NCBI Entrez System. The backbone LLM used by our system is Llama 2-Chat in its 7 billions parameter versions. We set to a low value the *temperature sampling* hyper-parameter that controls randomness of the model output at inference time. Temperature sampling is a modification of the softmax function in the last layer; a low temperature makes the output of the LLM more deterministic, thus favoring more probabilistic predictions. This conservatism is captured by the tendency of the model to produce more repetitive, focused and less diverse results based on the most commonly observed patterns in the training data [23]. In the context of diagnosis, it is preferable for the model to generate repetitive and poorly reasoned but correct responses, hence we use the value: temperature = 0.0001.

Also in favor of a more pronounced determinism, the *nucleus sampling* value top_p, is set to 0.95; i.e., only tokens having a combined probability of 95 % will be considered. We set the context window to 4096 tokens in order to preserve the precision characteristic of RAG. The framework used to implement the RAG technique is LangChain. Its purpose is

to enable developers to use other data sources and interact with other applications, enabling the construction of Chains. A chain takes care of concatenating a sequence of operations before bringing out the output. The simplest chain is one that generally takes an input prompt and generates an output. LangChain also provides several useful tools for splitting text into chunks and vectorization [24].

Embeddings are quantitative representations of textual units that capture their semantic essence in numerical form. When applied to similar textual content, these representations generate closely aligned vectors. The *embeddings model* used in this work is all-MiniLM-L6-v2, specifically a sentence transformer that maps sentences and paragraphs into a dense vector space of 384 dimensions. Once the vector representation has been obtained, we store it into the vector database Facebook AI Similarity Search (FAISS), which is part of a complete range of tools developed by Meta to enable the search for embeddings of multimedia documents.

B. Workflow

Figure 2 represents a complete workflow concerning differential diagnosis between Bipolar Disorder and Psychomotor Agitation, while Figure 3 shows the response of our method for differential diagnosis of ADHD and Autism spectrum disorder.

- 1) The physician is asked to enter the names of the two disorders. The terms are used as keywords for searching abstracts on the PubMed database through the Entrez APIs. The returned articles are sorted by relevance and

DISORDERS

- ❖ ADHD
- ❖ Autism spectrum disorder

SCORE: 7/10

COMMENT: Based on the documents provided, there is significant overlap between Attention Deficit Hyperactivity Disorder (ADHD) and Autism Spectrum Disorder (ASD). Both disorders share common symptoms and characteristics, including difficulties in social interaction, communication and repetitive behaviour.

According to the document entitled [**Autism Spectrum Disorder and Attention Deficit/Hyperactivity Disorder: Challenges in Diagnosis and Treatment**], the coexistence of ADHD and ASD is a common phenomenon and poses unique challenges for diagnosis and treatment. The paper states that the two disorders share several functional brain alterations, and the established treatment for ADHD is also effective in this comorbidity. Similarly, the paper entitled [**Medical Comorbidities in Children and Adolescents with Autism Spectrum Disorders and Attention Deficit/Hyperactivity Disorders: A Systemic Review**] notes that somatic disorders occur more often in adult psychiatric patients than in the general adult population, but the association between CHD and ASD is unclear due to a lack of integration of existing data. The authors emphasise the importance of initiating therapeutic interventions as soon as a diagnosis of autism is suspected, rather than waiting for a definitive diagnosis. The paper entitled [**Practical Considerations for the Assessment and Management of Attention Deficit/Hyperactivity Disorder (ADHD) in Adults**] highlights the similarities between ADHD and ASD in terms of their neuroevolutionary origins and the persistence of symptoms into adulthood. The authors note that although ADHD is typically considered a childhood disorder, its impact on adult functioning cannot be underestimated, especially when dealing with comorbid conditions such as anxiety and depression.

Fig. 3. Response of our method for differential diagnosis of ADHD and Autism spectrum disorder

chronology. The title and abstract of the results are extracted, verifying that the disease and symptom has at least one co-occurrence in the text. An example of concatenated abstracts can be seen Figure 2 in the box associated with the article extraction step.

- 2) Titles and abstracts are concatenated into a single text and split into chunks using Langchain's RecursiveCharacterTextSplitter with a size of 1024 characters and an overlap of 128.
- 3) The chunks are converted into their embeddings by the sentence transformer all-MiniLM-L6-v2 and stored within the FAISS vector database. The database is queried to compare the cosine similarity between the query vector and the vectors inside it.
- 4) A template prompt is carefully crafted, in which the k documents most similar to the user input are inserted. This phase is crucial because in addition to the context, instructions on how our model should behave must be inserted in the template. For this reason, we specify that we require a high degree of stringency to the provided context. We also specify the desired behavior in case the model does not have enough information to answer the question. We adopt this solution with the objective of avoiding hallucination by preventing the LLM from trying to answer the question even when required information is missing. The prompt template used in our system is as follows:

Prompt: *You are an expert medical assistant in charge of helping differential diagnosis, comparing diseases that show common symptoms. Your response must be based exclusively on the context provided below. Do not make up any information. Where possible, cite the exact titles or references of the sources used, and the title from the documents in the context. If the*

information is not found in the context or the context is empty, simply state: "I don't have enough information to answer the question. Assign a similarity rating from 0 (completely different) to 10 (synonymous) to the two disorders. Don't invent articles and use the title found in the context. Context: {context} Question: {question}"

- 5) Create the chain that will query the vector database and insert the most similar results into the prepared token. In our system, we can specify both the number of desired documents and the minimum similarity threshold that these documents must meet.
- 6) The user interacts with the system through a dialogue interface, where the model is invoked via LangChain to generate responses. A particularly useful feature of the system is its ability to display the documents utilized in generating the output tokens, providing transparency and traceability. For example, in the final box of the diagram, a response to a query about symptoms related to Bipolar Disorder is shown. Additionally, a similarity rating is provided. This rating is determined based on the factual information presented in the referenced articles, as well as the number of symptoms shared between the queried diseases.

IV. EXPERIMENTAL EVALUATION

To evaluate the effectiveness of the proposed system, we conducted an experimental study involving expert feedback from 12 professional psychologists (1 male, 11 female), aged 30 to 53. All participants were in their third year of the postgraduate program in Health Psychology at Cagliari University. The experts completed a total of 24 questionnaires, each designed to compare responses generated by three distinct systems (labeled A, B, and C) in supporting the differential diagnosis of 12 unique cases within the psychology domain. The 12 differential diagnosis cases were chosen with the help

of a psychologist which did not participate to annotations. The cases were provided to the annotators on a random basis, and questionnaires were acquired anonymously. In the questionnaires, the identities of the three evaluated systems were anonymized to mitigate potential bias in the experts' evaluations:

- 1) System A: Llama 2 with Knowledge Injection (**Llama 2 KI**) uses the Llama 2 7B model LLM with a standard mechanism of knowledge injection. Knowledge injection refers to the technique of inserting additional knowledge into the model prompt as is the case with the RAG technique. The difference is that there is no retrieval mechanism based on the cosine similarity of vectorized abstracts. The articles to be included are chosen using customized searches, which concern the type of article (journal paper), and keywords relevant to the disorders under consideration. Empty and pre-2020 articles are discarded, the first 500 abstracts are used for the extraction of common and different symptoms, and terminology is referenced to the International Classification of Diseases.
- 2) System B: **ChatGPT 4o**, which uses a plain ChatGPT 4o LLM.
- 3) System C: **Our method**; i.e., the novel solution introduced in this paper.

A. Experimental setup

The evaluation procedure was structured as follows. Each pair of disorders was analyzed by the three systems, which provided a similarity rating along with a written explanation of their symptoms similarities and differences. After reviewing each response, the psychologists completed a questionnaire containing a series of statements to which they indicated their level of agreement to the answers of the 3 systems using a 5-point Likert scale: 'Strongly Agree' (5), 'Agree' (4), 'Undecided' (3), 'Disagree' (2), and 'Strongly Disagree' (1).

The following statements were presented to the expert annotators.

- **S1**: The similarity score calculated by the system is accurate.
- **S2**: The description contains all relevant information to accurately distinguish the two diseases (e.g., main symptoms, key differences).
- **S3**: The descriptions are written in clear and easily understandable language.
- **S4**: The description is consistent and free of contradictions or ambiguities.
- **S5**: Descriptions highlight the most clinically relevant aspects (e.g., distinctive symptoms and points of overlap).
- **S6**: Details provided are useful for comparing diseases from a clinical perspective.

As can be seen from the list above, the evaluation criteria range from correctness and completeness of information to clarity of exposition.

Additionally, participants were asked to assign a numerical score from 1 to 10 to the disease pair to numerically measure

their similarity in terms of common symptoms, where 1 means 'completely different' and 10 means 'completely similar'.

Upon completing the evaluations for all the three systems, participants were asked to fill out a final questionnaire. This questionnaire required them to rank the three methods from best to worst, evaluate their perceived usefulness, and propose potential areas of application.

B. Inter-rater agreement

To evaluate the agreement among annotators, we used the Krippendorff's alpha (α) reliability coefficient, calculated as:

$$\alpha = 1 - \frac{D_o}{D_e}$$

where D_o represents the observed disagreement among values assigned to units of analysis, and D_e denotes the expected disagreement under the assumption that coding decisions are due to chance rather than the inherent properties of the units [25].

The coefficient ranges from -1 to 1, with values close to 1 indicating perfect agreement, values close to 0 indicating no agreement, and values close to -1 indicating systematic disagreement. The Krippendorff's alpha value computed on the 24 questionnaires was **0.313**, indicating a moderate level of agreement. The moderate agreement among raters can be attributed to several reasons. One key factor is subjectivity in clinical interpretation due to the unique clinical experience, theoretical orientation, or personal preferences of the psychologists. Additionally, differences in specialization could influence how therapists assess the responses in specific cases. Finally, differential diagnosis often involves subjective reasoning and probabilistic judgments, particularly when comparing closely related disorders, that can amplify rating variability.

C. Overall preferences of the systems

In order to evaluate the systems according to overall preference, the questionnaire asked participants to identify the best method. Our method received the most votes as the best option (13 out of 24) out of the three proposed solutions. We also applied the pairwise comparison method, which involves ranking the systems based on direct pairwise preferences, with one option deemed superior to another using a specific criterion. Here, the criterion was the number of times a system was preferred over another system. The results are as follows:

Llama 2 KI = 0; ChatGPT 4o = 1; Our method = 2.

This outcome indicates that our method was preferred over both the other methods, while ChatGPT 4o was preferred over Llama 2 KI. From these results, the following preference chain emerges: Llama 2 KI < ChatGPT 4o < Our method.

Finally, preferences were acquired using the Borda count method [26], where the least preferred method receives 1 point, the penultimate receives 2 points, and so on. The results obtained are as follows:

Llama 2 KI: 43; ChatGPT 4o: 49; Our method: 52.

	S1	S2	S3	S4	S5	S6	SIMIL.
Llama 2 KI	3.08	2.91	3.96	3.71	3.08	2.83	1.45
ChatGPT 4o	3.25	2.96	4.00	3.92	3.37	3.08	2.00
Our method	3.21	3.09	3.62	3.54	3.21	3.08	1.95

TABLE I

TABLE OF AVERAGE RATINGS FOR EACH STATEMENT S1 TO S6 (DESCRIBED IN SECTION IV-A); A LARGER VALUE INDICATES A MORE AGREEABLE VOTE. SIMIL. (SIMILARITY) IS THE AVERAGE OF THE DIFFERENCES BETWEEN THE SIMILARITY SCORE OF THE EXPERT ANNOTATORS AND THE ONE COMPUTED BY EACH SYSTEM: A LOWER VALUE INDICATES A MORE AGREEABLE VOTE. THE BEST RESULTS ACHIEVED FOR EACH SCORE ARE HIGHLIGHTED IN BOLD.

It can be observed that our method remains the most preferred approach, followed by ChatGPT 4o.

D. Rating of statements and similarity

We now specifically examine the individual methods, based on the questionnaires completed according to each method's response. Table I presents the average scores obtained by each method for each statement. The values that contributed to these averages correspond to the numerical conversion of the opinions expressed by selecting the corresponding box in the questionnaire. For each method, the values were summed, and divided by 24 (the number of questionnaires). The difference between the grade assigned by the tester and the grade given by the system is presented in the same table under the column titled "SIMIL."

From these results, all three systems provide clear and coherent written descriptions. Regarding clarity (S3) and lack of contradictions (S4), ChatGPT-4 achieves the best result, suggesting that more advanced LLMs excel at formulating unambiguous responses in natural language. However, some skepticism remains concerning the content of the descriptions. While all systems scored above average, there was no clear consensus indicating that the answers were deemed complete and useful for comparing the two disorders. Specifically, statements 2 and 6 recorded the lowest overall averages, underscoring issues with the completeness and utility of the responses for diagnostic purposes.

Concerning the assignment of a similarity score, the three systems are capable of generally identifying when two disorders are related, based on overlapping symptoms, but often fail to capture the finer nuances needed to assign an accurate score. The Similarity column reveals that an average error of up to 2 points still results in a positive overall classification. In this task, Llama 2 KI outperforms both ChatGPT 4o and our method, likely due to the use of entity recognition techniques that isolate symptoms and enable more accurate comparisons between the two disorders.

Regarding the utility of the system for differential diagnosis, responses to S3 indicate that our method outperforms both Llama 2 KI and ChatGPT 4o in providing descriptions that include all relevant information necessary to accurately distinguish between the two disorders, such as key differences and main characteristics. While ChatGPT 4o achieves the best results in providing clinically relevant aspects (S5), both our

method and ChatGPT 4o offer the most useful details for comparing diseases from a clinical perspective (S6).

E. Possible application areas

In the final questionnaire, the raters were asked to provide their opinion on the application of these solutions. The majority of psychologists, 14 out of 24, indicated that they agree with their use, with an average score of 3.41 out of 5 for this metric. Regarding its intended purpose, there was strong agreement on its potential to assist in diagnosis, followed by its use for research purposes. In contrast, its application as a therapy aid was clearly in the minority. The results related to the most useful application areas are reported in Figure 4.

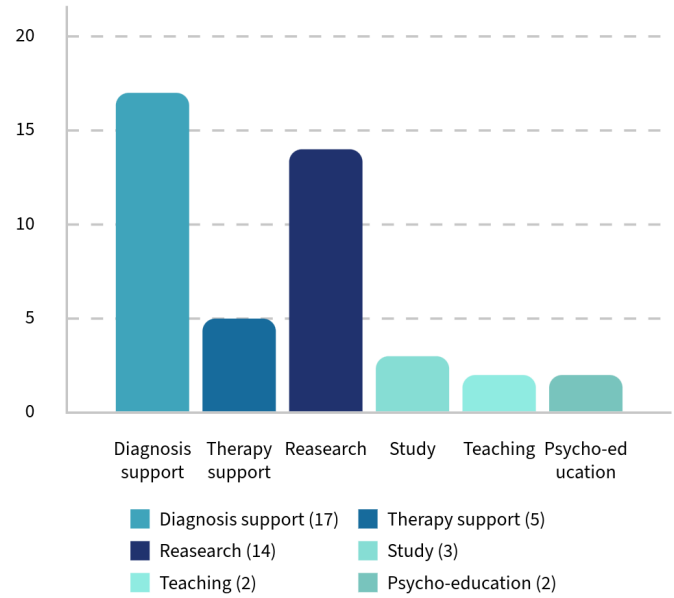


Fig. 4. Chart showing the number of votes for each area of use.

F. Discussion

Summarizing the opinions from the questionnaires, our method emerged as the preferred, particularly when compared to a system that relies on a simpler knowledge injection technique (Llama 2 KI). When compared to ChatGPT 4o, the gap narrows, largely due to the superior linguistic capabilities of the latter. A weak point of using a pure LLM is that it will always attempt to generate an answer, even in the absence of sufficient information, presenting the output in a convincing manner. On the other hand, our method integrated with a strict prompt and appropriate parameters, provides answers that are highly dependent on the available context, limiting linguistic variability in favor of greater adherence to the selected scientific literature. A key strength of our method is its ability to provide the practitioner with an explanation of the source of the presented information, allowing users to explore the sources in greater depth. This aligns with the feedback from the questionnaire, which emphasized that the role of the professional remains crucial and cannot be fully replaced by an automatic AI system.

V. CONCLUSION

In this paper, we presented a novel AI agent to support differential diagnosis through a combination of RAG and LLMs. Our work addresses the mistrust issue in the usage of AI, particularly generative models, in intelligent health-care systems, mainly due to unclear or unreliable sources of information. We experimentally compared our system with both a powerful LLM, and with a LLM using a standard knowledge injection method. The results of an evaluation with 12 professional therapists show that our system compares favorably with state-of-the-art LLMs in delivering the critical information required to differentiate between two diseases, with valuable details for supporting the differential diagnosis.

Several issues need to be addressed in future work. The quality of the data used is crucial: including only common articles may not significantly enhance a domain expert's knowledge, but incorporating a broader contexts may reduce response accuracy. A refined method to carefully select relevant literature to distinguish among similar disorders should be devised. To address this issue, a more comprehensive context needs to be considered, possibly combining PubMed resources with other data. Electronic Medical Records could enhance the system, though privacy concerns hinder their integration.

Further improvements in RAG could include document re-ranking, query rewriting, and better chunking strategies for embeddings. Moreover, fine-tuning, particularly with domain-specific models like MedPALM 2 and BioBERT, would allow for additional system's inferences, supporting the model in deriving more complex medical deductions. Finally, future evaluations should consider a larger trial and reproduce full diagnostic interactions to test the system's capabilities in real-world scenarios.

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REFERENCES

- [1] G. Briganti and O. Le Moine, "Artificial intelligence in medicine: today and tomorrow," *Frontiers in medicine*, vol. 7, p. 509744, 2020.
- [2] N. Abbasi and H. K. Hussain, "Integration of artificial intelligence and smart technology: Ai-driven robotics in surgery: Precision and efficiency," *Journal of Artificial Intelligence General science (JAIGS) ISSN: 3006-4023*, vol. 5, no. 1, pp. 381–390, 2024.
- [3] P. Hamet and J. Tremblay, "Artificial intelligence in medicine," *metabolism*, vol. 69, pp. S36–S40, 2017.
- [4] E. Khodabandehloo and D. Riboni, "Collaborative trajectory mining in smart-homes to support early diagnosis of cognitive decline," *IEEE Transactions on Emerging Topics in Computing*, vol. 9, no. 3, pp. 1194–1205, 2020.
- [5] M. Haag, C. Igel, M. R. Fischer, G. M. E. S. GMA *et al.*, "Digital teaching and digital medicine: A national initiative is needed," *GMS journal for medical education*, vol. 35, no. 3, 2018.
- [6] A. J. Thirunavukarasu, D. S. J. Ting, K. Elangovan, L. Gutierrez, T. F. Tan, and D. S. W. Ting, "Large language models in medicine," *Nature medicine*, vol. 29, no. 8, pp. 1930–1940, 2023.
- [7] X. Yu, Z. Zhang, F. Niu, X. Hu, X. Xia, and J. Grundy, "What makes a high-quality training dataset for large language models: A practitioners' perspective," in *Proceedings of the 39th IEEE/ACM International Conference on Automated Software Engineering*, 2024, pp. 656–668.
- [8] G. Perković, A. Drobnyak, and I. Botički, "Hallucinations in llms: Understanding and addressing challenges," in *2024 47th MIPRO ICT and Electronics Convention (MIPRO)*. IEEE, 2024, pp. 2084–2088.
- [9] E. Ullah, A. Parwani, M. M. Baig, and R. Singh, "Challenges and barriers of using large language models (llm) such as chatgpt for diagnostic medicine with a focus on digital pathology—a recent scoping review," *Diagnostic pathology*, vol. 19, no. 1, p. 43, 2024.
- [10] E. Khodabandehloo, D. Riboni, and A. Alimohammadi, "Healthxai: Collaborative and explainable ai for supporting early diagnosis of cognitive decline," *Future Generation Computer Systems*, vol. 116, pp. 168–189, 2021.
- [11] K. Singhal, T. Tu, J. Gottweis, R. Sayres, E. Wulczyn, L. Hou, K. Clark, S. Pföhl, H. Cole-Lewis, D. Neal *et al.*, "Towards expert-level medical question answering with large language models," *arXiv preprint arXiv:2305.09617*, 2023.
- [12] J. Lee, W. Yoon, S. Kim, D. Kim, S. Kim, C. H. So, and J. Kang, "Biobert: a pre-trained biomedical language representation model for biomedical text mining," *Bioinformatics*, vol. 36, no. 4, pp. 1234–1240, 2020.
- [13] Y. Gao, Y. Xiong, X. Gao, K. Jia, J. Pan, Y. Bi, Y. Dai, J. Sun, M. Wang, and H. Wang, "Retrieval-augmented generation for large language models: A survey," *arXiv preprint arXiv:2312.10997*, 2023.
- [14] H. Soudani, E. Kanoulas, and F. Hasibi, "Fine tuning vs. retrieval augmented generation for less popular knowledge," *arXiv preprint arXiv:2403.01432*, 2024.
- [15] H. Touvron, L. Martin, K. Stone, P. Albert, A. Almahairi, Y. Babaei, N. Bashlykov, S. Batra, P. Bhargava, S. Bhosale *et al.*, "Llama 2: Open foundation and fine-tuned chat models," *arXiv preprint arXiv:2307.09288*, 2023.
- [16] C. Wu, W. Lin, X. Zhang, Y. Zhang, W. Xie, and Y. Wang, "Pmc-llama: toward building open-source language models for medicine," *Journal of the American Medical Informatics Association*, p. ocae045, 2024.
- [17] A. Johnson, L. Bulgarelli, T. Pollard, S. Horng, L. A. Celi, and R. Mark, "Mimic-iv," *PhysioNet*. Available online at: [https://physionet.org/content/mimiciv/1.0/\(accessed August 23, 2021\)](https://physionet.org/content/mimiciv/1.0/(accessed August 23, 2021)), pp. 49–55, 2020.
- [18] D. Lamba, W. H. Hsu, and M. Alsadhan, "Predictive analytics and machine learning for medical informatics: A survey of tasks and techniques," in *Machine learning, big data, and IoT for medical informatics*. Elsevier, 2021, pp. 1–35.
- [19] Z. Lu, "Pubmed and beyond: a survey of web tools for searching biomedical literature," *Database*, vol. 2011, p. baq036, 2011.
- [20] Q. Jin, R. Leaman, and Z. Lu, "Pubmed and beyond: biomedical literature search in the age of artificial intelligence," *EBioMedicine*, vol. 100, 2024.
- [21] P. Lewis, E. Perez, A. Piktus, F. Petroni, V. Karpukhin, N. Goyal, H. Küttler, M. Lewis, W.-t. Yih, T. Rocktäschel *et al.*, "Retrieval-augmented generation for knowledge-intensive nlp tasks," *Advances in Neural Information Processing Systems*, vol. 33, pp. 9459–9474, 2020.
- [22] S. Wang, W. Zhou, and C. Jiang, "A survey of word embeddings based on deep learning," *Computing*, vol. 102, no. 3, pp. 717–740, 2020.
- [23] M. Renze and E. Guven, "The effect of sampling temperature on problem solving in large language models," *arXiv preprint arXiv:2402.05201*, 2024.
- [24] O. Topsakal and T. C. Akinci, "Creating large language model applications utilizing langchain: A primer on developing llm apps fast," in *International Conference on Applied Engineering and Natural Sciences*, vol. 1, no. 1, 2023, pp. 1050–1056.
- [25] K. Krippendorff, "Agreement and information in the reliability of coding," *Communication methods and measures*, vol. 5, no. 2, pp. 93–112, 2011.
- [26] P. Emerson, "The original borda count and partial voting," *Social Choice and Welfare*, vol. 40, no. 2, pp. 353–358, 2013.